

Packet Scheduling Algorithms (WFQ and FCFS)

1. Objective

The objective of this lab is to implement, simulate, and compare two packet scheduling algorithms: Weighted Fair Queuing (WFQ) and First-Come-First-Serve (FCFS). The simulation should use a multithreaded architecture to emulate concurrent packet generation and scheduling. Multiple traffic sources will share a single output link with finite buffer capacity.

2. System Description

The system consists of N independent traffic sources, each generating packets at a given rate with variable lengths and weights. Each source has parameters: P_i (packet rate), L_i^{min}, L_i^{max} (packet size range), w_i (weight for WFQ), and t_{bi}, t_{ei} (start and end times). The link capacity C defines the rate at which packets are transmitted. The shared queue can hold a limited number of packets. If the buffer is full, the system drops packets according to WFQ (smallest finish number) or FCFS (earliest or new arrival as per your design).

3. Input Format

The program will read simulation parameters from an input file. The input file structure is as follows:

First line:

`N=<num_sources> T=<simulation_time> C=<link_capacity> B=<queue_capacity>`

Then N lines, one per source:

`P_i L_min,i L_max,i w_i t_bi t_ei`

These parameters define the rate, packet size range, weight, and active duration for each source.

4. Simulation Input Sets

Input Set A ($N = 4$)

N=4 T=1200 C=100000 B=150
15 800 1600 5 0.05 0.85
25 600 1400 3 0.03 0.80
30 900 1700 2 0.02 0.90
60 1200 2000 1 0.01 0.95

Input Set B ($N = 4$)

N=4 T=1000 C=80000 B=120
20 500 1500 4 0.00 0.80
35 700 1400 3 0.10 0.85
25 900 1800 2 0.05 0.90
50 1000 2000 1 0.15 0.95

5. Expected Outputs

Your program should compute and record the following system-level performance metrics for both WFQ and FCFS:

1. Server Utilization – average link usage over time.
2. Fairness Index – based on Jain’s Fairness Index formula.
3. Average Packet Delay – mean time between generation and transmission.
4. Packet Drop Probability – fraction of packets dropped among total generated.

6. Report and Analysis

Prepare a concise report that includes:

1. Overview of WFQ and FCFS algorithms.
2. Simulation setup (number of sources, link capacity, buffer size).
3. Results – metrics collected for both input sets and algorithms.
4. Comparative discussion:

- How do WFQ and FCFS differ in fairness, delay, and drop probability?
- Why does WFQ generally achieve better fairness than FCFS?
- Identify cases where FCFS may perform better.

5. Conclusion summarizing key observations.

7. Submission

Submit a single compressed folder containing:

- Source code and build instructions.
- Input and output files for both algorithms.
- Technical report (PDF) including performance analysis and conclusions.

Note :- Use either C,C++ or Java